Environmental Product Declaration

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 and ISO 21930:2017 for:

Heat-Treated Medium and Heavy Plates

from

Beijing Shougang co.,Ltd

Programme:	The International EPD System, www.environdec.com
Programme operator:	EPD International AB
Type of EPD:	EPD of multiple products from a company, based on the average results of the product group.
EPD registration number:	EPD-IES-0025361:001
Version date:	2025-09-03
Validity date:	2030-09-02

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com

GENERAL INFORMATION

Programme Information	
Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
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Product Category Rules (PCR)
CEN standard EN 15804 serves as the Core Product Category Rules (PCR)
Product Category Rules (PCR): PCR 2019:14 Construction products, version 2.0.1 (valid until 2030-04-07), CEN standard EN 15804 serves as the core PCR. ***UN CPC code: 412.***
PCR review was conducted by: Rob Rouwette (chair) and Noa Meron (co-chair) who can be contacted via support@environdec.com.

Third-party Verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006, via: ☒ **Individual EPD verification without a pre-verified LCA/EPD tool** Third-party individual verifier: *Matthew Fishwick from Fishwick Environmental Ltd.* Approved by: International EPD System
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.
For further information about comparability, see EN 15804 and ISO 14025.

INFORMATION ABOUT EPD OWNER

Owner of the EPD: Beijing Shougang co.,Ltd

Address: No. 99 Shijingshan Road, Shijingshan District, Beijing, China

Contact: Zhangling Wang (mobile: 15832557303)

Address and contact information of the LCA practitioner commissioned by the EPD owner: Shuhan.huang@ivl.se, Beijing City, China.

Description of the organisation: Beijing Shougang Co., Ltd. (referred to as "Shougang") is the highest quality core asset in Shougang Group's steel industry and is the integration platform for Shougang Group's steel and upstream iron ore businesses in China. Jingtang Company is the largest production base of Shougang Group's steel sector, with products including automotive plates, electrical steel, hot-rolled and pickled plates, and medium heavy plates.

Jingtang Company has always adhered to the concept of circular economy and green low-carbon development. In recent years, Jingtang Company has been listed as one of the first pilot enterprises for creating resource-saving and environmentally responsible enterprises in the steel industry in China, one of the first pilot units for carbon emission management system construction in Hebei Province, a typical case of "waste free enterprise" by the Ministry of Industry and Information Technology, and has been awarded multiple titles such as "Green Factory", "Water Efficiency Leader", "Green Development Benchmark Enterprise", "Dual Carbon Best Practice Energy Efficiency Benchmark Demonstration Factory", etc.

Jingtang Company continues to carry out pollution control and emission reduction, environmental protection improvement and renovation work, continuously increasing investment in environmental protection governance, and achieving emission standards throughout the entire process. After being rated as an A-level enterprise by the Hebei Department of Ecology and Environment in November 2020, it passed the new A-level evaluation in Hebei Province in December 2024, making a green transformation demonstration for the industry.

Product-related or management system-related certifications: ISO 9001:2015 Quality Management System Certification, ISO 14001:2015 Environmental Management Systems, ISO 45001:2018 Occupational Health and Safety Management Systems, ISO/IATF 16949:2016 automotive profile quality systems, CNAS Laboratory Accreditation Certificate.

PRODUCT INFORMATION

Product name: Heat-Treated Medium and Heavy Plates

Product identification: EN10225 :2009 Weldable structural steels for fixed offshore structures - Technical delivery conditions

UN CPC code: 412

Product description: The thickness of the medium heavy plate products of Shougang is 5-380mm, and the maximum width of the finished product is 4100mm. With the first 400mm thick slab continuous casting machine in China with the technology of voestalpine, the maximum thickness of the non-destructive testing qualified steel plate can be produced up to 252mm. The products are widely used in

major engineering projects and equipment manufacturing at home and abroad, such as shipbuilding and offshore engineering, wind power generation, gas pipeline, petroleum refining, building structures, energy reserves, and engineering machinery. The European standard products have passed CE certification, covering S235-S960 levels, with certificate number 0035-CPR-A206, which can meet the relevant market demand. The products list and chemical composition can be found below:

Table 1 Product list and their chemical compositions

European Norm	Product included in the group			
EN10025-3*	S275NL, S355J2+N, S355N, S355NL, S355NL+Z25, S355NL+Z35			
EN10025-6*	S550Q, S690Q			
Element	**unit**	**Average**	**Highest**	**Lowest**
C	%	0.14	0.20	0.04
Si	%	0.25	0.80	0.20
Mn	%	1.47	1.70	0.90
P	%	0.02	0.03	0.00
S	%	0.01	0.02	0.00
N	%	0.00	0.01	0.00
Al	%	0.03	0.06	0.02
Nb	%	0.02	0.06	0.00
Ti	%	0.01	0.05	0.00
Ni	%	0.12	4.00	0.00
Cr	%	0.04	1.50	0.00
Mo	%	0.04	0.70	0.00
Fe	%	0.00	0.01	0.00

*EN 10025-3: Hot rolled products of structural steels Part3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels
*EN 10025-6: Hot rolled products of structural steels Part6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition

The manufacturing process of Heat-Treated Medium and Heavy Plates begins with the production of pellet ore and sinter ore, progressing through blast furnace ironmaking, converter steelmaking, refining, and continuous casting to produce high-quality slabs. These slabs are then subjected to rolling and heat treatment to form the final medium and heavy plate products. Following quality inspection, the plates are packaged, preparing them for shipment to customers.

Name and location of production site: Shougang Jingtang United Iron & Steel Co., Ltd.(Steel and Electric Power Park, Caofeidian Industrial Zone, Tangshan City, Hebei Province, China).

CONTENT DECLARATION

The mass (weight) of one unit of a product, as purchased or per declared unit: 1 000 kg.

The result of this study is calculated based on the average results of the product group. The content declaration below is in reference to average content of the product group.

Table 2 Content declaration

Product content	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic materials		
			Mass-% of product	kg C / declared unit	kg CO_2 / declared unit
Steel slab	1.0E+3	2.1%	0.0%	0.0	0.0
TOTAL	1.0E+3	2.1%	0.0%	0.0	0.0

Table 3 Content declaration of packaging

Packaging materials	Mass, kg	Mass-% (versus the product)	Biogenic material, kg C/ declared unit	Biogenic material, kg CO_2/ declared unit
Timber dunnage (wood)	6.0E-02	0.0060%	3.5E-02	1.3E-01
TOTAL	6.0E-02	0.0060%	3.5E-02	1.3E-01

1 kg biogenic carbon in the product/packaging is equivalent to the uptake of 44/12 kg of CO_2.

For construction product EPDs compliant with EN 15804, the content declaration shall at least declare the substances contained in the product that are listed in the “Candidate List of Substances of Very High Concern for Authorization”, in case their content exceeds the limits for registration with the European Chemicals Agency (0.1% of the weight of the product). Shougang declares that their products covered by the study do not contain substances of very high concern (SVHC) as defined and listed in the European Chemicals Agency (ECHA) Candidate List of substances of very high concern for Authorization, in levels above 0.01% by weight for the products

LCA INFORMATION

Declared unit: 1,000 kg of Heat-Treated Medium and Heavy Plates (based on average product results).

Time representativeness: January 2024 to December 2024.

Geographical scope: A1-A3: China; A5: Global, C1-C4: Global, D: Global.

Database(s) and LCA software used: Software: SimaPro 9.6.0.1, database: ecoinvent 3.10.1 (system boundary cut-off).

Description of system boundaries: (b) Cradle-to-gate with options, modules C1-C4, module D and optional modules A5.

Process flow diagram: Process flow diagram of the product system, divided into the life-cycle stages and modules, showing the main processes included and the system boundary of the LCA.

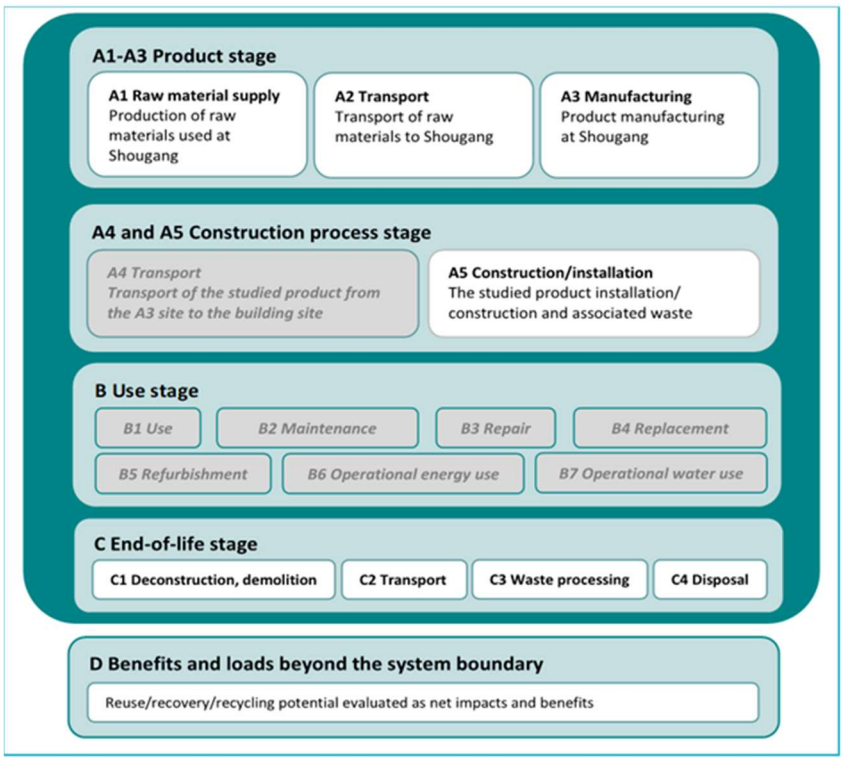

Figure 1 System diagram

More information:

Scenarios

- Module A5: Since lacking supporting information, the environmental impact during the construction stage of the product is not considered. However, since the package contains biogenic carbon, depending on the recommendations of the PCR, it is assumed that the packaging becomes waste in this stage and is treated to end-of-life stage. So, the treatment of the packaging with the biogenic carbon dioxide is considered in this stage.
- Module C1-C4 :The modeling of Module C was based on Section 4.8.4 of the PCR. The declared scenario aligns with the World Steel Association's guidelines and represents a mix of recycling (85%) and landfill (15%). Corresponding scenarios for 100% recycling and 100% landfill have also been declared. The declared scenario (S0) serves as the base case, while the 100% recycling and 100% landfill scenarios (S2) are considered alternative scenarios.
- The C1 module is assumed that 1.1 kWh diesel would be used to demolition/deconstruction one tonne of steel. After C1, it is assumed that 15% of the product would be sent to C4 for landfill, and 85% of steel for recycling according to The World Steel Association (S0). For the C2 module, the assumption has been made that all waste products would be transported for 80 km by truck. For C3, it is assumed that processes such as loading and unloading at the sorting facility (diesel consumption), mechanical sorting (electricity consumption), and fragging (diesel consumption) are required. Additionally, it is assumed that 5% of the waste product is lost during processing. For C4, it assumed that compacting of inert construction waste for landfills would consume diesel.
- Module D: Since each scenario for Modules C3 or C4 corresponds to a scenario in Module D, the waste product recycling rates are set at 85% for S0, 100% for S1, and 0% for S2. To prevent double counting of benefits, secondary materials are excluded from the benefit calculation. Specifically, for S0, 807.5 kg of steel exits Module C3 and enters Module D. Given the recycled content of 2.1%, the net flow considered in Module D is 790.7 kg. For S1 and S2, the net flows considered are 930.2 kg and 0.0 kg, respectively. Beside, the benefit (credit) could be generated through applying the

waste produce to replace the new one. For the studied product, it could be seen as the steel scrap after deconstruction. After sorting and fragging, it can be added to the production of new steel products, replacing the liquid iron or pig iron.

Allocation

- In the studied system, there were co-products produced along with the studied product. For co-products with low economic value, such as ashes and slags, conservative assumptions were made, as the effort of allocation is disproportionate to any improvement in accuracy. For energy-related co-products and steel scrap (excluding internal recycling), economic allocation methods were applied.
- The allocation of waste in this study follows the polluter-pays principle that is made operational according to the below described method, which is also called cut-off approach. The system boundary to the subsequent product system is set where the waste (e.g., the discarded product) reaches the end-of-waste state, i.e., when it has become a usable flow (e.g., for reuse, recycling and/or energy recovery). At the system boundary, cut-off allocation is applied, i.e., all unit processes before the point of end-of-waste shall be assigned to the product system generating the waste and all unit processes after the point of end-of-waste shall be assigned to the subsequent product system. Besides, where third parties pay for waste transportation, the end-of-waste state is reached before the transport.
- Steel scrap used as inputs are classified into 4 types: closed-loop scrap, pre-consumer scrap (different production line), purchased pre-consumer scrap, and purchased post-consumer scrap. For closed-loop scrap (and also some closed-loop dust and slag), as they are recycled internally in the same product system, thereby no co-products leave the product system, and no allocation is necessary. For pre-consumer scrap (different production line) and purchased pre-consumer scrap, economic allocation methodology was implemented using customer-provided price. And the ratio of the pre-consumed scrap price to the steel slab (60.7%) is used as the allocation factor.
- The declared unit corresponds to the average weight of the products. For this data, allocation based on physical properties was applied. Specifically, the energy use, auxiliary materials consumption, and waste generation per declared unit are calculated by dividing the usage by the total amount of the products.

Cut-off

The cut-off criteria established by the PCR is that data for elementary flows to and from the product system contributing to a minimum of 95% of the declared environmental impacts shall be included (not including processes that are explicitly outside the system boundary). No cut-off for the studied product has been applied in the study.

Information on the modelling of infrastructure/capital goods

The production, maintenance, and after-use treatment of capital goods, such as machines, factories, etc., “overhead” activities, such as heating of buildings and lighting, and the activities of the employees are not included in the life cycle. In this study, the infrastructure and capital goods are not included in the LCA analysis since they are used many times for several years for the product manufacturing. According to the PCR, it should be excluded.

Additional technical information:

- The climate impact of the electricity of A3 (GWP-GHG indicator) is 0.426 kg CO_2 eq./kWh. Shougang’s factory has an independent power supply network, which includes electricity purchased from the national grid mix, third-party photovoltaic power, third-party power generated from dry quenching in the coking process, and other self-generated electricity (such as electricity recovered from surplus energy in production stages). For the national grid mix, because China is relatively large and the electricity generation structure varies from province to province, the

electricity data for China requires the use of a sub-national electricity mix according to the PCR requirements. Therefore, the electricity mix from North China Grid(NCGC) was used, and it is region where the plant is located. This NCGC power mix includes hydroelectric power, wind power, natural gas power, petroleum power, nuclear power and hard coal power. Note that the Guarantees of Origin market in China represents an extremely small proportion of production and consumption, and therefore the consumption mix is effectively the same as the residual mix.

- The climate impact of the biogas of A3 (GWP-GHG indicator) is 0.0263 kg CO_2 eq./kWh.
- The LCA analysis applied the EN 15804 as the calculation methodology. All results in this LCA analysis are calculated by EN 15804+A2. The "EN 15804 reference package" is calculated based on EF 3.1.

Table 4 Modules declared, geographical scope, share of primary data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	**A1**	**A2**	**A3**	**A4**	**A5**	**B1**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**C1**	**C2**	**C3**	**C4**	**D**
Modules declared	X	X	X	ND	X	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	CN	CN	CN	-	GLO	-	-	-	-	-	-	-	GLO	GLO	GLO	GLO	GLO
Share of primary data	34.27%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	ND (single site)			-	-	-	-	-	-	-	-	-	-	-	-	-	-

Data quality

A summary of the data quality assessment, in line with requirements of PCR in Section 4.6.4. is listed below. The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

Table 5 The declaration of data sources, reference years, data categories, and share of primary data

Process	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Production of coke	Database	ecoinvent 3.10.1	2023	Secondary data	0.00%
Production of coal	Database	ecoinvent 3.10.1	2023	Secondary data	0.00%
Generation of electricity	Collected	EPD owner	2024	Primary data	3.78%
Production of nitrogen	Collected	EPD owner	2024	Primary data	20.61%
Generation of the fuels	Collected	EPD owner	2024	Primary data	4.56%
Generation of the biogas	Collected	EPD owner	2024	Primary data	0.00%
Transportation of raw materials and packaging	Collected	EPD owner	2024	Primary data	5.32%
Emission and waste treatment	Collected	EPD owner	2024	Primary data	0.00%
Other process	Database	ecoinvent 3.10.1	2023	Secondary data	0.00%
Total share of primary data, of GWP-GHG results for A1-A3					34.27%

The EPD covers Heat-treated Medium and Heavy Plates from Shougang's factory in Hebei Province, China, which provided data for the period 2024-01-01 to 2024-12-31. The product is manufactured through a complete steelmaking process, including blast furnace ironmaking, converter steelmaking, refining, continuous casting, rolling and heat treatment. Due to its wide range of applications, it is assumed that disposal of packaging waste in global, and the product's end-of-life stage is also in global. Background data was sourced from the ecoinvent 3.10.1. The data quality assessment is based on EN15804 Annex E Table E-1 (Data quality level and criteria of the Global Guidance Principles for Life Cycle Assessment Databases (UN Environment 2011). In general, time representation of the dataset's selection is very good for the studied product, the technical representation is good, the geographical representation is good. No fair, poor or very poor data was found during the assessment of relevant data.

ENVIRONMENTAL PERFORMANCE

LCA results of the products - main environmental performance results

Mandatory impact category indicators according to EN 15804

Results per 1 000 kg of Heat-Treated Medium and Heavy Plates								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-fossil	kg CO_2 eq.	2.5E+03	1.2E-03	5.6E-01	1.6E+01	5.4E+00	1.4E+00	-1.3E+03
GWP-biogenic	kg CO_2 eq.	6.5E-01	1.3E-01	1.5E-03	2.7E-03	1.1E-02	7.4E-04	-1.5E-01
GWP-luluc	kg CO_2 eq.	6.2E-01	5.8E-07	4.3E-04	6.2E-03	5.0E-03	7.4E-04	-3.0E-01
GWP-total	kg CO_2 eq.	2.5E+03	1.3E-01	5.6E-01	1.6E+01	5.4E+00	1.4E+00	-1.3E+03
ODP	kg CFC 11 eq.	6.5E-05	2.5E-11	7.2E-09	2.2E-07	6.1E-08	3.7E-08	-3.7E-06
AP	mol H^+ eq.	1.9E+01	6.5E-06	4.3E-03	5.2E-02	3.7E-02	9.8E-03	-4.4E+00
EP-freshwater	kg P eq.	7.3E-01	5.6E-07	7.6E-05	1.2E-03	1.2E-03	1.2E-04	-3.4E-01
EP-marine	kg N eq.	3.3E+00	2.3E-05	1.7E-03	1.7E-02	1.3E-02	3.7E-03	-1.0E+00
EP-terrestrial	mol N eq.	3.5E+01	2.6E-05	1.8E-02	1.8E-01	1.4E-01	4.1E-02	-1.1E+01
POCP	kg NMVOC eq.	9.6E+00	1.1E-05	5.4E-03	7.3E-02	4.3E-02	1.4E-02	-3.7E+00
ADP-minerals&metals*	kg Sb eq.	7.8E-03	2.9E-09	3.4E-06	5.0E-05	3.2E-05	2.8E-06	-6.7E-04
ADP-fossil*	MJ	4.0E+04	2.2E-02	7.0E+00	2.2E+02	6.7E+01	3.2E+01	-1.2E+04
WDP*	m^3	1.4E+03	-9.5E-03	4.5E-02	9.9E-01	5.5E-01	1.3E+00	-8.0E+01
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption							

** Disclaimer:*
The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator.
The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks."
The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3)."

Additional mandatory and voluntary impact category indicators

Results per 1 000 kg of Heat-Treated Medium and Heavy Plates								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[1]	kg CO_2 eq.	2.5E+03	4.9E-03	5.6E-01	1.6E+01	5.4E+00	1.4E+00	-1.3E+03

Resource use indicators

Results per 1 000 kg of Heat-Treated Medium and Heavy Plates								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	1.2E+03	7.2E-01	6.4E-01	2.8E+00	7.0E+00	4.5E-01	-2.4E+02
PERM	MJ	7.2E-01	-7.2E-01	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
PERT	MJ	1.2E+03	3.1E-04	6.4E-01	2.8E+00	7.0E+00	4.5E-01	-2.4E+02
PENRE	MJ	3.9E+04	2.2E-02	7.0E+00	2.2E+02	3.7E+02	8.5E+01	-1.2E+04
PENRM	MJ	3.6E+02	0.0E+00	0.0E+00	0.0E+00	-3.0E+02	-5.4E+01	0.0E+00
PENRT	MJ	4.0E+04	2.2E-02	7.0E+00	2.2E+02	6.7E+01	3.2E+01	-1.2E+04
SM	kg	2.3E+01	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
RSF	MJ	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
NRSF	MJ	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
FW	m^3	3.4E+01	-2.2E-04	1.4E-03	2.9E-02	1.9E-02	3.1E-02	-2.6E+00
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water							

[1] This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO_2 is set to zero.

Waste indicators

Results per 1 000 kg of Heat-Treated Medium and Heavy Plates								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	2.5E+02	5.5E-07	1.7E-04	5.5E-03	2.1E-03	4.6E-04	-2.3E-01
Non-hazardous waste disposed	kg	2.4E+02	6.1E-02	6.2E-02	1.0E+01	4.9E-01	1.9E+02	-6.1E+01
Radioactive waste disposed	kg	1.4E-02	5.2E-09	2.2E-06	4.4E-05	6.2E-05	5.2E-06	-2.3E-03

Output flow indicators

Results per 1 000 kg of Heat-Treated Medium and Heavy Plates								
Indicator	**Unit**	**A1-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Material for recycling	kg	0.0E+00	0.0E+00	0.0E+00	0.0E+00	8.1E+02	0.0E+00	0.0E+00
Materials for energy recovery	kg	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Exported energy, electricity	MJ	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00
Exported energy, thermal	MJ	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00	0.0E+00

Additional LCA results (other environmental performance results) of the products

Results for additional scenarios for modules C1-C4 and D:The most representative scenario (for the geographical scope of the EPD) has been declared as the main environmental performance results. And another two scenarios:100% recycling (S1) and 100% landfill (S2) are declared here.

Indicator	Unit	C1 S1	C2 S1	C3 S1	C4 S1	D S1	C1 S2	C2 S2	C3 S2	C4 S2	D S2
GWP-fossil	kg CO_2 eq.	5.6E-01	1.6E+01	6.3E+00	3.6E-01	-1.5E+03	5.6E-01	1.6E+01	0.0E+00	7.1E+00	0.0E+00
GWP-biogenic	kg CO_2 eq.	1.5E-03	2.7E-03	1.3E-02	1.9E-04	-1.8E-01	1.5E-03	2.7E-03	0.0E+00	3.8E-03	0.0E+00
GWP-luluc	kg CO_2 eq.	4.3E-04	6.2E-03	5.8E-03	1.9E-04	-3.5E-01	4.3E-04	6.2E-03	0.0E+00	3.8E-03	0.0E+00
GWP-total	kg CO_2 eq.	5.6E-01	1.6E+01	6.3E+00	3.6E-01	-1.5E+03	5.6E-01	1.6E+01	0.0E+00	7.1E+00	0.0E+00
ODP	kg CFC 11 eq.	7.2E-09	2.2E-07	7.2E-08	9.6E-09	-4.4E-06	7.2E-09	2.2E-07	0.0E+00	1.9E-07	0.0E+00
AP	mol H^+ eq.	4.3E-03	5.2E-02	4.4E-02	2.5E-03	-5.2E+00	4.3E-03	5.2E-02	0.0E+00	5.0E-02	0.0E+00
EP-freshwater	kg P eq.	7.6E-05	1.2E-03	1.4E-03	3.2E-05	-4.0E-01	7.6E-05	1.2E-03	0.0E+00	6.3E-04	0.0E+00
EP-marine	kg N eq.	1.7E-03	1.7E-02	1.5E-02	9.7E-04	-1.2E+00	1.7E-03	1.7E-02	0.0E+00	1.9E-02	0.0E+00
EP-terrestrial	mol N eq.	1.8E-02	1.8E-01	1.7E-01	1.1E-02	-1.3E+01	1.8E-02	1.8E-01	0.0E+00	2.1E-01	0.0E+00
POCP	kg NMVOC eq.	5.4E-03	7.3E-02	5.0E-02	3.7E-03	-4.4E+00	5.4E-03	7.3E-02	0.0E+00	7.4E-02	0.0E+00
ADP-minerals&metals*	kg Sb eq.	3.4E-06	5.0E-05	3.8E-05	7.5E-07	-7.8E-04	3.4E-06	5.0E-05	0.0E+00	1.5E-05	0.0E+00
ADP-fossil*	MJ	7.0E+00	2.2E+02	7.9E+01	8.2E+00	-1.5E+04	7.0E+00	2.2E+02	0.0E+00	1.6E+02	0.0E+00
WDP*	m^3	4.5E-02	9.9E-01	6.5E-01	3.4E-01	-9.5E+01	4.5E-02	9.9E-01	0.0E+00	6.8E+00	0.0E+00

ABBREVIATIONS

Abbreviation	Definition
General Abbreviations	
EN	European Norm (Standard)
EPD	Environmental Product Declaration
EF	Environmental Footprint
GPI	General Programme Instructions
ISO	International Organization for Standardization
LCA	Life Cycle Assessment
PCR	Product Category Rules
c-PCR	Complementary Product Category Rules
CEN	European Committee for Standardization
CLC	Co-location centre
CPC	Central product classification
GHS	Globally harmonized system of classification and labelling of chemicals
GRI	Global Reporting Initiative
Environmental Impact Indicators (EN 15804)	
GHG	Greenhouse gas
GWP	Global Warming Potential (kg CO_2 eq.)
GWP-fossil	Global Warming Potential from fossil sources (kg CO_2 eq.)
GWP-biogenic	Global Warming Potential from biogenic sources (kg CO_2 eq.)
GWP-luluc	Global Warming Potential from land use and land use change (kg CO_2 eq.)
GWP-total	Total Global Warming Potential (kg CO_2 eq.)
GWP-GHG	Global Warming Potential for greenhouse gases (kg CO_2 eq.)
ODP	Ozone Depletion Potential (kg CFC-11 eq.)
AP	Acidification Potential (mol H^+ eq.)
EP	Eutrophication Potential
EP-freshwater	Freshwater eutrophication potential (kg P eq.)
EP-marine	Marine eutrophication potential (kg N eq.)
EP-terrestrial	Terrestrial eutrophication potential (mol N eq.)
POCP	Photochemical Ozone Creation Potential (kg NMVOC eq.)
ADP	Abiotic Depletion Potential
ADP-minerals&metals	Abiotic depletion potential for non-fossil resources (kg Sb eq.)
ADP-fossil	Abiotic depletion potential for fossil resources (MJ)
WDP	Water Deprivation Potential (m^3)
Resource Use Indicators	
PERE	Use of renewable primary energy excluding renewable primary energy resources used as raw materials (MJ)
PERM	Use of renewable primary energy resources used as raw materials (MJ)
PERT	Total use of renewable primary energy resources (MJ)
PENRE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials (MJ)

PENRM	Use of non-renewable primary energy resources used as raw materials (MJ)
PENRT	Total use of non-renewable primary energy resources (MJ)
SM	Use of secondary material (kg)
RSF	Use of renewable secondary fuels (MJ)
NRSF	Use of non-renewable secondary fuels (MJ)
FW	Use of net fresh water (m^3)
Waste Indicators	
HW	Hazardous Waste (disposed) (kg)
NHW	Non-Hazardous Waste (disposed) (kg)
RW	Radioactive Waste (disposed) (kg)
Output Flow Indicators	
CFR	Components for Reuse (kg)
MR	Material for Recycling (kg)
MER	Materials for Energy Recovery (kg)
EEE	Exported Energy, Electricity (MJ)
EET	Exported Energy, Thermal (MJ)
Lifecycle Stages / Modules	
A1	Raw material supply
A2	Transport
A3	Manufacturing
A5	Construction/Installation
Other Relevant Terms	
SVHC	Substances of Very High Concern
EC No.	European Community Number
CAS No.	Chemical Abstracts Service Number
MJ	Megajoule
kg	Kilogram
m^3	Cubic Meter
NMVOC	Non-Methane Volatile Organic Compounds
Sb eq.	Antimony Equivalents
P eq.	Phosphorus Equivalents
N eq.	Nitrogen Equivalents
CFC-11 eq.	Chlorofluorocarbon-11 Equivalents
CO_2 eq.	Carbon Dioxide Equivalents
kg C	Kilograms of Carbon
kg CO_2 eq.	Kilograms of Carbon Dioxide Equivalent
ND	Not Declared

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VERSION HISTORY

Original Version of the EPD, 2025-09-03

EPD
INTERNATIONAL EPD SYSTEM
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